

Deep Learning Analysis of Smartphone and Digital Stethoscope Phonocardiograms for
Detection of Reduced Left Ventricular Ejection Fraction

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A thesis submitted in partial fulfillment of the requirements for
the Degree of Bachelor of Biomedical Sciences at
the University of Hong Kong

24 April 2026

Declaration

I, Wonjun Yang, a student of Bachelor of Biomedical Sciences at the University of Hong Kong, hereby declare that the work presented in this final year project titled “Deep Learning Analysis of Smartphone and Digital Stethoscope Phonocardiograms for Detection of Reduced Left Ventricular Ejection Fraction” is entirely my own and has been completed under the guidance of Professor Joshua Ho.

I affirm that all the research, analysis, and findings included in this project are original and have not been plagiarised or copied from any other source. Any references, quotations, or ideas derived from external sources have been duly acknowledged and cited in accordance with academic standards.

I also declare that this project has not been submitted in part or full for any other degree, diploma, or certification at this or any other institution.



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24 April 2026

Acknowledgements

I would like to express my sincere gratitude to Professor Joshua Ho for supervising this project and for broadening my view of cardiovascular disease research beyond its technical aspects. I am especially thankful that he gave me the opportunity to work on this topic, because I truly enjoyed it.

I am also very grateful to Dr Chi Yan Ooi for taking the time to explain the study design and the process of heart sound data collection. His generosity in sharing that background gave me a much clearer understanding of the clinical setting and practical context of this work.

I would also like to thank Chutong Xiao, a PhD student, for first introducing me to the dataset and for helping me understand its structure and background at the beginning of the project.

I am likewise thankful to Professor Chun-Ka Wong, whose clinical study provided the de-identified dataset that made this dissertation possible.

Above all, I am grateful to my family for their love and unwavering support throughout my undergraduate years. There were many times when I felt lost and unsure of what I wanted to do in the future, but they never made me feel rushed to have everything figured out. They gave me reassurance and encouragement to keep growing, keep searching, and keep moving forward, even when the path ahead was not yet clear.

In the vastness of space and the immensity of time, it feels remarkable that I have had the chance to meet people who guide me, encourage me, and quietly shape the course of my life. I have been very fortunate to receive such kindness, generosity, and support. This dissertation is only a small piece of work, but it carries my heartfelt gratitude to all those who have helped me find my way.

Abstract

Reduced left ventricular ejection fraction (LVEF) is clinically important, but its detection often relies on echocardiography, limiting screening. Phonocardiograms (PCGs) can be recorded using smartphones and digital stethoscopes, making them attractive for lower-cost screening. However, reduced LVEF is a functional target, not a directly audible abnormality. It is also unclear whether PCG-based deep learning performs consistently across different recording devices. This study evaluated whether reduced LVEF could be detected from PCGs recorded with smartphones and a digital stethoscope, and how performance changed across within-device, cross-device, and pooled-device settings. Data from 487 adult cardiology outpatients, comprising 5,822 recordings, were analysed. Mel-frequency cepstral coefficients (MFCCs) and gammatonegrams were compared across six pretrained deep learning backbones. Performance was assessed using the F1 score, sensitivity, specificity, and the area under the receiver operating characteristic curve (AUROC). Among the six backbones tested, SwinV2-Tiny was used in the best-performing within-device configuration for all recording devices, but the preferred time-frequency representation differed by device. Final within-device AUROC values were similar across devices (0.664-0.666). Cross-device AUROC values were less stable (0.542-0.672), suggesting device-related domain shift. On the pooled test set, the pooled-device model achieved the strongest performance (AUROC 0.680; F1 29.5%), suggesting that training on recordings from heterogeneous devices may improve robustness. Pooled-device performance varied by auscultation site, with pulmonic and aortic sites best. These results indicate that PCGs alone may contain information relevant to reduced-LVEF screening. A direct comparison of within-device, cross-device, and pooled-device evaluation for PCG-based reduced-LVEF detection across smartphones and a digital stethoscope has not been reported in prior published studies. This study therefore highlights device-related domain shift as a central methodological issue for PCG-based reduced-LVEF screening. Broader validation across additional devices, populations, and recording conditions, together with methods that address domain shift, remains a key next step.

Keywords

Reduced left ventricular ejection fraction; phonocardiogram; smartphone auscultation; deep learning; domain shift

Abbreviations

AI	artificial intelligence
AUPRC	area under the precision-recall curve
AUROC	area under the receiver operating characteristic curve
CNN	convolutional neural network
ECG	electrocardiogram
LVEF	left ventricular ejection fraction
MFCC	mel-frequency cepstral coefficient
PCG	phonocardiogram

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1. Introduction

1.1. Clinical background and relevance of reduced left ventricular ejection fraction

Heart failure is a common and serious cardiovascular syndrome, covering a broad range of left ventricular ejection fraction states [1,2]. One clinically important subgroup is heart failure with reduced left ventricular ejection fraction (LVEF). LVEF is the proportion of blood pumped out of the left ventricle with each heartbeat. It is used in routine care as a practical marker of systolic pumping function. Current guidance continues to use LVEF of 40 percent or below as a clinically important threshold because it helps define reduced systolic function and guides treatment decisions [1,3,4].

Reduced LVEF is a meaningful screening target because it guides further cardiac assessment and treatments that have been shown to improve outcomes in systolic heart failure [1,3]. It helps identify patients who may need earlier investigation and treatment [1,3]. If it is not recognised early, there may be delays in echocardiography, specialist review, and treatment escalation. Table 1 summarises heart failure classification by LVEF.

Type of Heart Failure	Criteria
Heart failure with reduced ejection fraction (HFrEF)	LVEF $\leq$ 40%
Heart failure with improved ejection fraction (HFimpeEF)	Previous LVEF $\leq$ 40% and follow-up LVEF $>$ 40%
Heart failure with mildly reduced ejection fraction (HFmrEF)	LVEF 41% - 49% with evidence of increased left ventricular filling pressures
Heart failure with preserved ejection fraction (HFpEF)	LVEF $\geq$ 50% with evidence of increased left ventricular filling pressures

Table 1. Classification of heart failure by LVEF. Adapted from Heidenreich et al. [1].

Symptoms such as breathlessness, fatigue, and reduced exercise tolerance are often nonspecific and can arise from many conditions, not only reduced LVEF [3]. Echocardiography remains the reference investigation for quantifying LVEF and confirming structural or functional cardiac abnormalities [1,3]. However, echocardiography is relatively resource-intensive and not well suited to broad screening in every setting. Wider access depends on sufficient service capacity [5,6]. Hence, a low-cost screening approach could be valuable if it helps identify patients who should undergo echocardiography earlier.

Reduced LVEF is a more precise target than heart failure in general. Heart failure refers to a broad clinical syndrome caused by structural or functional cardiac abnormalities that lead to symptoms or signs [1,3,4]. Heart failure can occur across reduced, mildly reduced, and preserved ejection fraction states [1,3,4]. Reduced LVEF does not define all heart failure, but it is a measurable cardiac phenotype with a clear threshold [1,3,4]. This makes it easier to define consistently as a screening endpoint than heart failure as a broader syndrome. It also provides a practical basis for asking whether a lower-cost screening approach could help identify patients who should undergo echocardiography earlier.

Nevertheless, reduced LVEF is not directly audible in the same way as a murmur. Instead, ventricular dysfunction may alter cardiac timing and filling patterns, thereby influencing heart sounds [7-12]. Earlier studies using phonocardiography and acoustic cardiography found associations between third and fourth heart sounds, systolic time intervals, and left ventricular dysfunction [7-11]. Several of these studies were conducted in selected cardiac patient groups and often relied on combined sound and timing measurements rather than a routine screening workflow based solely on phonocardiograms (PCGs) [8-11]. These studies remain important because they support the physiological rationale for exploring heart sounds as an indirect marker of ventricular dysfunction [7-12]. At the same time, a PCG-only approach remains attractive because many earlier methods relied on additional measurements and more specialised instrumentation.

The key question is not whether heart sounds can replace echocardiography. The question is whether they can provide an earlier clue that reduced systolic function may be present, so that echocardiography can be considered earlier in patients whose symptoms do not clearly suggest reduced LVEF.

1.2. Challenges in earlier screening and the potential of phonocardiograms

1.2.1. Limitations of routine auscultation

Conventional cardiac auscultation refers to listening to the heart with a stethoscope placed on the chest. It is cheap, familiar, and non-invasive. Yet subtle acoustic signs are difficult to detect consistently in routine practice. Studies of gallop-sound detection found poor inter-observer agreement and limited accuracy, even among experienced clinicians [13,14]. Reduced LVEF is not expected to generate an obvious acoustic abnormality that can be recognised reliably during routine listening. If any sound pattern related to reduced LVEF is weak and indirect, routine listening alone is unlikely to support consistent screening.

Digital signal capture partly addresses this limitation by preserving the acoustic signal for repeated review and more standardised analysis [15-17]. Once a heart sound is recorded, it can be reviewed by multiple observers, transformed in reproducible ways, and analysed using the same processing pipeline across many patients. Digital capture shifts the task from one-time listening to reproducible signal analysis, which is one reason PCGs are attractive for machine learning studies.

Digital capture also changes what can be asked of heart sounds. Conventional auscultation is judged mainly by whether a single clinician can hear and interpret a sound at a given moment. Stored PCGs can instead be analysed as signals that may contain weak patterns across timing, frequency, and amplitude. For reduced-LVEF screening, the value of a model would not lie in detecting one obvious acoustic abnormality more accurately. It would lie in extracting subtle acoustic patterns more consistently across many recordings.

This broader clinical interest in low-cost point-of-care screening is reflected in a study that used an electrocardiogram (ECG)-enabled stethoscope to identify reduced LVEF during routine examination [18]. It shows that reduced-LVEF screening has already been tested within a routine examination workflow. However, that ECG-enabled stethoscope study does not address the PCG-only question because its screening signal is derived from single-lead ECG input, rather than heart sounds alone [18]. A PCG-only approach remains important to study because it asks whether screening information can be obtained from the heart sound alone.

Figure 1 shows the standard precordial auscultation areas and examples of pathological heart sounds. This figure helps place the problem in context because it shows clearer acoustic abnormalities that are more easily recognised during routine cardiac auscultation. Reduced LVEF is harder to study because it is a functional phenotype rather than one distinct pathological sound.

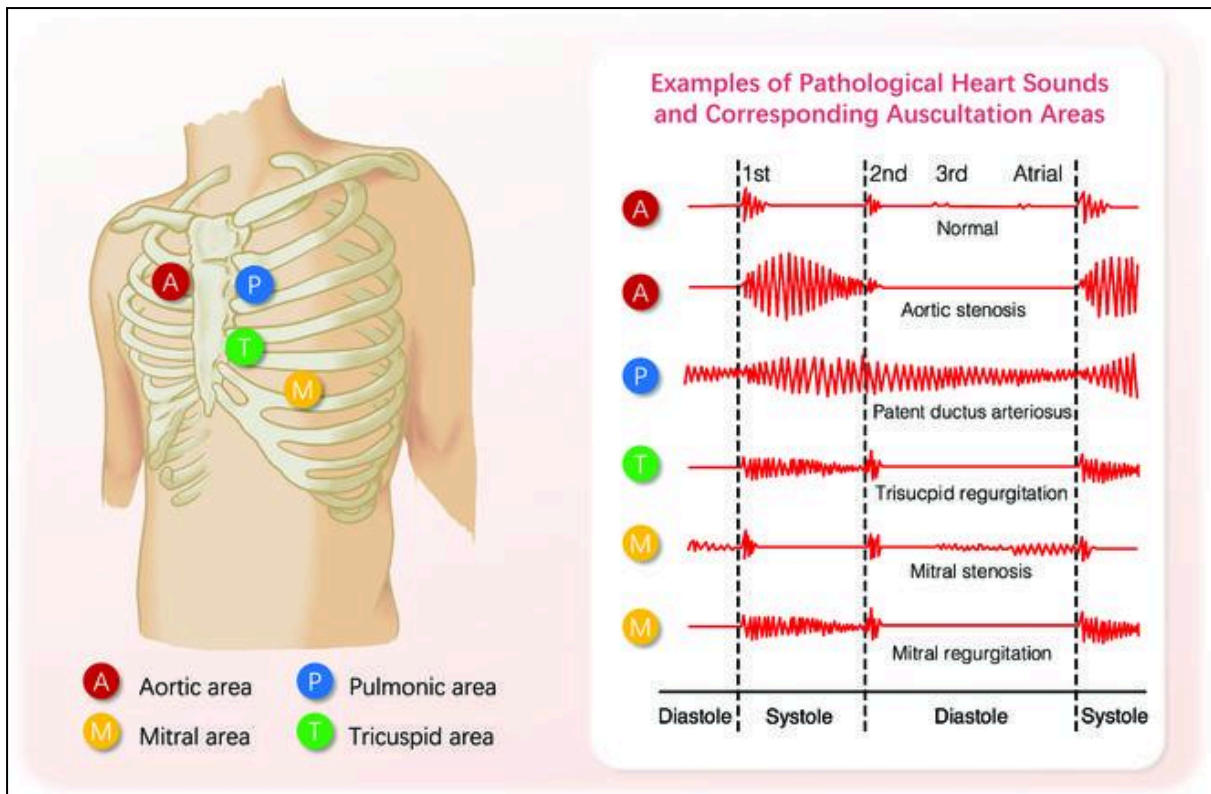


Figure 1. Standard precordial auscultation areas and examples of pathological heart sounds. Reproduced from Zhao et al. [16] under the Creative Commons Attribution-ShareAlike (CC BY-SA) 3.0 license.

1.2.2. Relevance of smartphones and digital stethoscopes

Stored PCGs are attractive for screening research because they require fewer resources than echocardiography [5,6]. They can be recorded repeatedly and incorporated into routine examination workflows [17,18]. This is especially relevant when recordings can be made with smartphones as well as digital stethoscopes. Studies using smartphones show that useful heart sound recordings can be captured on consumer devices and that signal processing can improve recording quality for later analysis [19-22].

The same studies also show that recording conditions matter. Recording quality is affected by placement, background noise, and other recording conditions [19-22]. Studies in both controlled and hospital settings show that feasibility does not necessarily mean that these sources of variation are removed [19,20]. Additionally, the smartphone literature mainly addresses whether usable recordings can be obtained and how recording quality is affected, rather than whether unsupervised home screening is realistic [19-22].

An intermediate outpatient setting between unsupervised self-recording and a highly controlled research dataset is also important to consider. Recordings collected by research personnel in routine outpatient care are more controlled than unsupervised self-recording, but they still resemble ordinary clinical practice more than a highly controlled research dataset. A

key question is whether such supervised outpatient recordings preserve enough information for reduced-LVEF discrimination.

Studying smartphones and digital stethoscopes together is not only about the cost. Smartphones are already widely available in clinical and research settings, whereas digital stethoscopes are designed specifically for cardiac sound capture [17]. These are not interchangeable recording devices, and differences in how they capture and process signals are part of the scientific problem [16,17]. A clinically useful screening approach may need to generalise across both types of hardware rather than remain limited to a single device ecosystem. The key question is whether recordings from different hardware platforms retain enough disease-related information for a machine learning model trained on recordings from one device to perform well on recordings from another device.

1.3. Prior heart sound deep learning and the challenge of detecting reduced left ventricular ejection fraction

1.3.1. Established applications of heart sound deep learning

Earlier automated PCG pipelines often relied on handcrafted descriptors or explicit feature engineering [15,16]. Deep learning approaches reduce reliance on a fixed set of manually chosen features by learning task-relevant patterns from transformed inputs. Recent reviews show that most deep learning literature on heart sounds focuses on murmur detection, structural heart disease, or congenital heart disease [15,16]. Many studies transform one-dimensional waveforms into time-frequency representations because heart sounds vary across both time and frequency, and these representations capture that variation in a form that deep learning models can use [15,16].

Most clinical successes in heart sound deep learning have been in the detection of murmurs and structural heart disease. Digital stethoscope studies have shown that deep learning can detect cardiac murmurs, including murmurs linked to structural heart disease, in real clinical settings [23,24]. They also show that heart sound models can work outside small, carefully selected datasets and can be integrated into a clinical examination workflow [23,24].

However, murmur studies should not be taken as direct evidence for reduced-LVEF screening. Their acoustic target is more directly present in the recorded sound. A murmur is a sound phenomenon with a clearer relationship to what is being recorded. Reduced LVEF is different. It is an echocardiographic measure of ventricular systolic function, so any related acoustic pattern is likely to be weaker, less specific, and more easily influenced by other physiological or recording-related factors. This difference in target type should be kept in mind when interpreting heart sound studies.

1.3.2. Reduced left ventricular ejection fraction as a more challenging functional target

Reduced LVEF is expected to be harder to detect from heart sounds than a murmur. A murmur is an audible acoustic event. Reduced LVEF is a quantitative measure of ventricular systolic function, so any related sound pattern is likely to be weaker and more indirect [7-11]. The target is also imperfectly measured because LVEF estimation itself is subject to inter-observer and inter-method variability [25]. This means that the target measurement is less precise, and the signal is subtler than in many murmur detection studies.

The difficulty is not only theoretical. Any weak sound pattern associated with reduced LVEF can be obscured by differences between patients, background noise, and ordinary variations in how the recording was made [7-11,19-22]. Even if ventricular dysfunction influences the heart sound, the relevant pattern may be small compared with other factors that also affect the recording [7-11,19-22]. Therefore, the choice of time-frequency representation matters. It determines which parts of the signal are preserved or compressed before the data

are given to the deep learning model. Reviews of heart sound deep learning support mel-frequency cepstral coefficients (MFCCs) and other representations as reasonable starting points, but they do not establish whether MFCCs, gammatone representations, or other image-like inputs are preferable for a subtle reduced-LVEF screening task across multiple recording devices [15,16]. MFCCs provide a compact spectral summary. Gammatone representations are more auditory-inspired time-frequency representations. Comparing them helps test whether a more compressed or a more detailed representation is more useful for this task.

Earlier phonocardiographic and acoustic cardiography studies show that ventricular dysfunction can affect recorded heart sounds [7-12]. However, these studies were done before current supervised deep learning workflows. They often relied on selected cardiac cohorts, and frequently combined heart sounds with timing features or simultaneous ECG information [8-12]. They do not establish a PCG-only screening workflow based on routine outpatient recordings.

A more recent PCG-only transfer learning study on diastolic dysfunction is closer to this present study because it treats heart sounds as inputs for a machine learning model that targets a functional cardiac abnormality rather than a murmur [26]. Diastolic dysfunction is also clinically important, but its echocardiographic assessment is more complex and does not rest on a single thresholded measure [27,28]. Hence, the study on diastolic dysfunction is relevant, though not directly comparable. It does not address reduced LVEF or the specific challenge of cross-device generalisation across heterogeneous recording domains [26].

The most clinically direct reduced-LVEF studies rely primarily on ECG-enabled or multimodal biosignal approaches [18,29,30]. The Lancet Digital Health study by Bachtiger and colleagues used a single-lead ECG recorded during ECG-enabled stethoscope examination to screen for reduced LVEF in routine care [18]. Zeng and colleagues reported a multimodal model that used synchronous ECG and PCG recordings from a digital stethoscope to identify left ventricular dysfunction [29]. Guo and colleagues then reported a large cohort validation of reduced-LVEF detection using an ECG-enabled digital stethoscope platform [30]. Overall, these studies show that reduced-LVEF screening based on recorded cardiac biosignals can be tested in patient cohorts rather than discussed solely as a theoretical possibility [18,29,30]. However, they do not isolate what PCGs alone contribute. Their reported performance cannot be assumed to transfer to a PCG-only model, as the signal source differs [18,29,30].

A digital stethoscope foundation model study is one of the closest studies to the PCG-only reduced-LVEF question because it included separate PCG-only, ECG-only, and combined reduced-LVEF models within one digital stethoscope ecosystem [31]. However, like the ECG-enabled and multimodal studies discussed above, it still operates within a single digital stethoscope platform. It does not explicitly test whether PCG-only performance remains strong when the recording domain changes across smartphones and a stethoscope. This leaves open the possibility that part of what the model learns reflects device-specific

sound characteristics and recording conditions rather than features that truly distinguish reduced LVEF.

Collectively, the aforementioned literature supports three conclusions. First, heart sounds are a plausible source of indirect information about ventricular dysfunction [7-12]. Second, deep learning of heart sounds has shown greater clinical success for targets with more direct acoustic features, especially murmurs and structural disease [15,16,23,24]. Third, reduced-LVEF screening from recorded cardiac biosignals is supported by previous studies [18,29-31]. However, the evidence is still stronger for ECG-based or multimodal approaches, and for single-ecosystem digital stethoscope studies, than for PCG-only reduced-LVEF work that directly tests whether performance remains reliable when the recording domain changes.

1.4. Device heterogeneity and domain shift as the main methodological gap

1.4.1. Importance of device-related domain shift

Device heterogeneity is not a minor technical issue for this task. Different recording devices can differ in microphone response, contact with the chest wall, and frequency characteristics [16,17]. Smartphone studies also show that placement, ambient noise, movement, and contact pressure affect recording quality [19-22]. Related domain-shift and transfer learning work suggests that generalisation across recording domains can be difficult [32,33]. When the disease signal is subtle, a model can end up learning device-related patterns instead of disease-related patterns [32,34]. This risk is especially important in reduced-LVEF screening because the expected acoustic signal is weaker than a classic murmur.

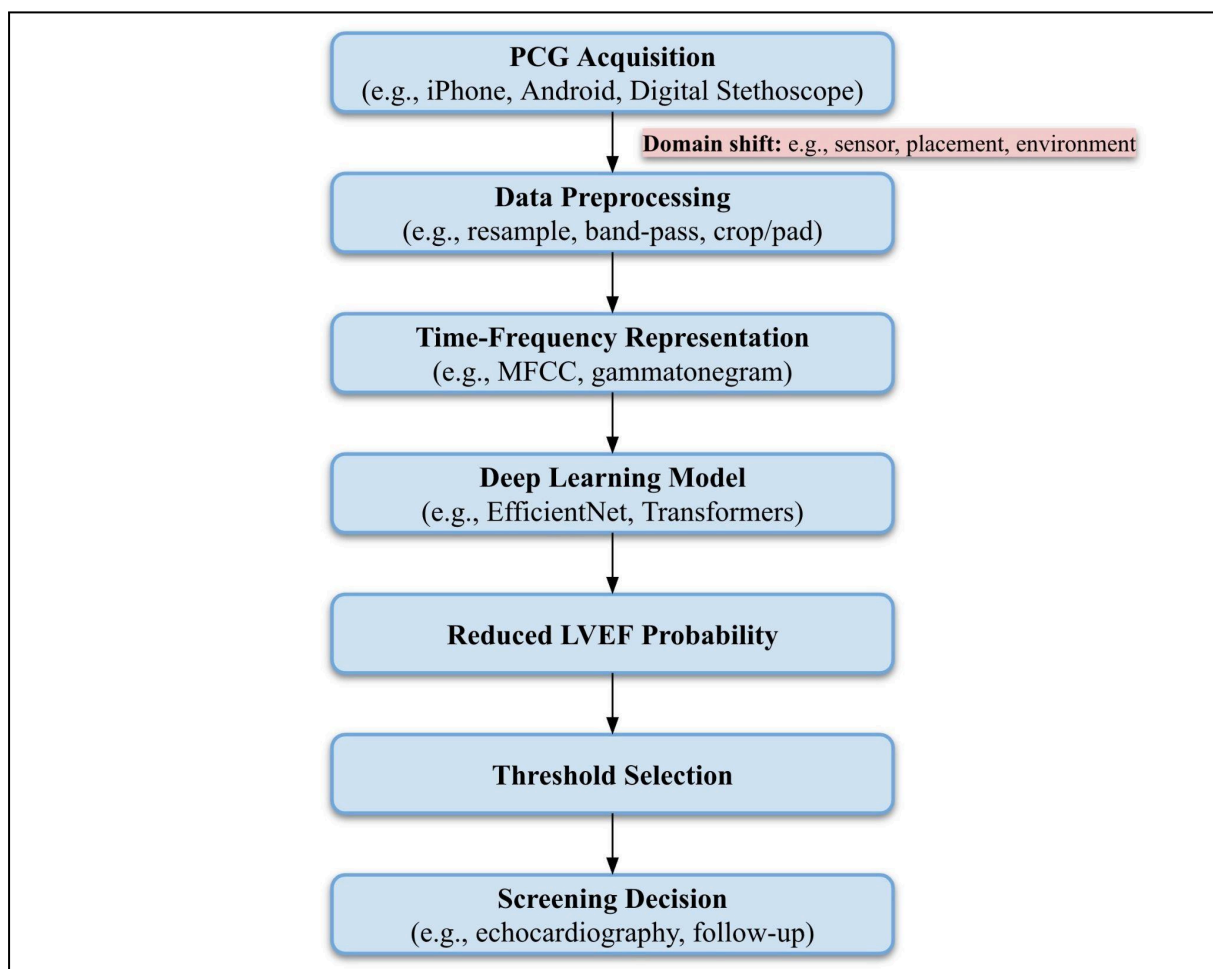


Figure 2. Sources of device-related domain shift in a PCG screening pipeline.

Several related PCG studies have tried to address this problem, but they involve other classification tasks [32-34]. Researchers have proposed learnable filterbanks, transfer learning, and robustness-focused preprocessing to improve performance across domains [32-34]. These studies are relevant, but they should not be overinterpreted. They focus on broader abnormality detection tasks in which the disease-related signal is likely stronger than

in reduced-LVEF screening. Their results suggest that reduced-LVEF screening may face similar transfer problems when recording conditions or hardware change.

Results from a single recording platform can therefore be misleading. Strong within-device performance can look persuasive, even if part of what the model learns is tied to one device's spectral signature or recording artefacts. For reduced-LVEF screening, that possibility is not a secondary issue. It is a central threat to interpretation. A strong result is clinically meaningful only if the model learns patterns related to ventricular function rather than device-specific patterns that are unrelated to the target phenotype.

1.4.2. Within-device, cross-device, and pooled-device evaluation settings

Each patient can contribute multiple recordings across devices and auscultation sites. This means that evaluation design directly affects model generalisation. If recordings from the same patient appear in both model development and held-out evaluation, performance can appear stronger than it really is [35]. Patient-level separation is therefore essential when the aim is to test generalisability rather than memorisation of repeated recordings.

Within-device evaluation means training and testing on the same device type. Cross-device evaluation means training on one device and testing on a different device without additional retraining. Pooled-device evaluation means training on recordings from all device groups together and then evaluating on mixed-device held-out data. These settings are complementary, not interchangeable. Within-device results examine whether reduced-LVEF information can be learned within a single recording domain. Cross-device results ask whether that information transfers when the recording domain changes. Pooled-device results ask whether broader exposure to device variation improves robustness, rather than only improving performance within a single device type.

Each setting leads to a different conclusion. Strong within-device performance may suggest that a useful reduced-LVEF signal is present, but it does not indicate whether that signal is separable from device-specific features. Weak cross-device performance may indicate that the learned representation depends too heavily on one acquisition domain. Strong pooled-device performance would suggest something different, because it would indicate that at least part of the disease-related signal can still be recovered when the model is exposed to broader device variation during training. Reporting only one setting can give an incomplete and possibly misleading picture of clinical usefulness.

Another question is which model type is most suitable once the evaluation setting is defined properly. Related PCG studies compare several time-frequency representations and both convolutional and transformer-based models, but they do not identify one combination that is consistently best [15,16,36,37]. For a subtle functional target recorded across multiple devices, the choice of representation and model family is part of the scientific question. In this study, MobileNet and EfficientNet represented lightweight convolutional models,

whereas SwinTransformer represented hierarchical transformer models. Convolutional models are useful for learning local patterns, whereas hierarchical transformers can model broader relationships across the input representation [37,38]. Comparing both model families helped test whether reduced-LVEF information depends more on local acoustic details or on wider patterns. Smaller and larger model variants were also included to examine the effect of model size. Model size matters because larger models are not always more useful if they become more sensitive to device-specific patterns instead of the underlying cardiac signal.

The unresolved issue is not just which representations and models to use. It also concerns how performance should be evaluated. In reduced-LVEF screening, the positive class is often much less common than the negative class [18,29,30]. For that reason, no single evaluation metric captures practical usefulness on its own [39-41]. Precision-recall analysis can be informative when performance on the positive class is a main concern [39]. However, broader guidance on clinical artificial intelligence evaluation recommends interpreting several metrics together [40], and recent methodological work argues that the area under the precision-recall curve (AUPRC) is not always better than the area under the receiver operating characteristic curve (AUROC) under class imbalance [41]. In this study, AUROC was kept as a summary measure, because related reduced-LVEF and stethoscope screening studies commonly report AUROC [18,30,31]. Its interpretation is clearer alongside the F1 score, sensitivity, and specificity, which make screening trade-offs more explicit.

In summary, emerging reduced-LVEF evidence from studies conducted within a single digital stethoscope platform does not address the PCG-only cross-device question for screening across smartphones and digital stethoscopes [31]. The unresolved issue is not simply whether heart sounds can carry reduced-LVEF information, but whether that information remains usable when recording domain changes. This methodological gap provides the main rationale for this study. It also leaves open the future possibility of smartphone-based pre-screening, although that application was not directly tested in this study. Future work should test whether performance remains reliable across additional devices, different populations, and more routine recording conditions.

1.5. Study aims and objectives

This study aimed to evaluate whether deep learning models trained on PCGs recorded with smartphones and a digital stethoscope could detect reduced LVEF. It used PCG inputs alone so that any predictive signal could be attributed directly to the heart sound rather than to other patient variables. More specifically, it examined how model performance changed when training and evaluation were performed within the same recording domain, across different recording domains, or on pooled multi-device data. Given the expected domain shift across recording devices, cross-device performance was expected to be less stable than within-device performance. Pooled-device training was expected to improve robustness by exposing models to a wider range of recording conditions during development. The objectives were as follows:

- I. To compare MFCC and gammatone representations across six deep learning backbones spanning lightweight convolutional neural networks (CNNs) and hierarchical vision transformers, and to identify the best-performing within-device configuration for iPhone, Android phone, and digital stethoscope recordings.
- II. To assess cross-device generalisation by evaluating each best-performing within-device model on held-out recordings from the other two devices.
- III. To determine whether a pooled-device model trained on recordings from all devices improved performance on a shared pooled test set relative to the best-performing within-device models.
- IV. To quantify model performance using the F1 score, sensitivity, specificity, and AUROC, and to explore whether pooled-device performance differed by auscultation site.

2. Methods

2.1. Study design and data source

This study investigated whether reduced LVEF could be detected from PCGs recorded with smartphones and a digital stethoscope. The analysis considered within-device, cross-device, and pooled-device experiments. The aim was to identify models that performed well within a single recording domain and to determine how performance changed across recording domains.

The heart sound recordings and LVEF values analysed in this study were derived from a de-identified stratified random subset of an observational clinical study registered on ClinicalTrials.gov as NCT06070298 (<https://clinicaltrials.gov/study/NCT06070298>). This parent clinical study was approved by the Institutional Review Board of the University of Hong Kong / Hospital Authority Hong Kong West Cluster (HKU/HA HKW IRB approval no. UW 23-436). The parent study focused on smartphone-based phonocardiography for murmur detection, whereas the present study used the same recordings to address a different task, namely, reduced-LVEF detection. Participants were adult cardiology outpatients who had undergone echocardiography within the preceding three years and provided consent during routine clinical care. According to the parent study protocol, individuals with implanted active medical devices in the torso, such as pacemakers and defibrillators, were excluded.

Recordings were collected by research personnel during routine clinical appointments and were not self-recorded. Each participant was recorded at the four standard cardiac auscultation sites, namely aortic, mitral, pulmonic, and tricuspid. iPhone recordings were obtained with an iPhone SE (3rd generation). Android recordings were obtained with either a Samsung Galaxy S22 Ultra or a Samsung Galaxy A54 5G. The de-identified dataset did not specify which Android phone was used for each individual recording. Digital stethoscope recordings were obtained with a 3M™ Littmann® CORE Digital Stethoscope. Recordings were obtained in routine clinical settings rather than acoustically controlled environments, and the order in which the devices were used was not randomised. After excluding 15 patients with missing LVEF values, the analysed cohort comprised 487 patients and 5,822 recordings in total. Most patients contributed 12 recordings, corresponding to four auscultation sites across three recording domains, although a small number contributed fewer recordings. The study cohort is summarised in Table 2.

Attribute	Details
Total patients	487
Total recordings	5,822
Patients by class	66 reduced LVEF (13.6%) 421 non-reduced LVEF (86.4%)
Recordings by class	789 reduced LVEF (13.6%) 5,033 non-reduced LVEF (86.4%)
Recordings by recording domain	iPhone: 1,948 Android phone: 1,938 Digital stethoscope: 1,936
Recordings by auscultation site	Aortic: 1,458 Mitral: 1,443 Pulmonic: 1,461 Tricuspid: 1,460
Recordings per patient	10-12 (466 of 487 patients have 12 recordings)
Duration per recording	7.5 s for all recordings
Audio format	WAV, mono, 4 kHz sampling rate

Table 2. Summary of the study cohort before data preprocessing.

2.2. Study cohort and outcome definition

The de-identified subset provided for this study consisted of WAV heart sound recordings and a CSV file containing echocardiographic LVEF values. No demographic or clinical covariates were available in this de-identified dataset. As a result, all models were trained solely on PCG inputs in this study.

Reduced LVEF was defined as an ejection fraction of 40% or below. This binary definition was used consistently throughout metadata construction, data splitting, model training, and evaluation. Patients with missing LVEF values were excluded prior to metadata construction, resulting in the exclusion of 15 patients from the de-identified subset.

The final cohort contained 66 patients with reduced LVEF and 421 patients without reduced LVEF. At the recording level, this corresponded to 789 reduced-LVEF recordings and 5,033 non-reduced-LVEF recordings. This class imbalance motivated the use of the positive class F1 score as the primary metric in this study.

2.3. Metadata construction

A metadata table was generated by linking each WAV recording to its corresponding ejection fraction, recording device, and auscultation site. This step produced one metadata row per recording. Splitting was later performed at the patient level to avoid leakage, yet the main analytical unit for model training and prediction was the individual recording. This design enabled comparisons across devices and auscultation sites.

2.4. Data preprocessing

All recordings were mono WAV files with an original sampling rate of 4 kHz and a duration of 7.5 seconds. Each waveform was first resampled from 4 kHz to 2 kHz to reduce computational cost, a choice that remained compatible with the frequency range retained later in preprocessing. Similar downsampling to 2 kHz has been used in recent PCG preprocessing pipelines [37,42]. A band-pass filter from 20 Hz to 800 Hz, a frequency range commonly used for heart sound analysis [43,44], was then applied to suppress very low-frequency drift and higher-frequency noise. After filtering, each recording was centre-cropped to 4 seconds to reduce computational cost during model training.

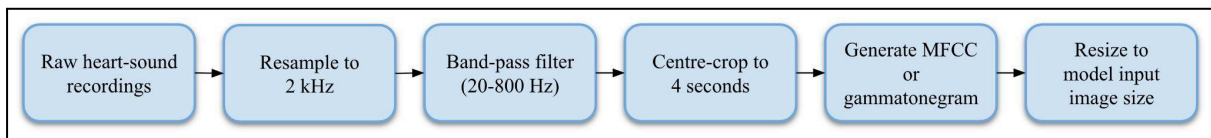


Figure 3. Data preprocessing workflow. Each recording was standardised before model input by resampling, band-pass filtering, centre-cropping, time-frequency transformation, and image resizing.

Two time-frequency representations were evaluated. The first was an MFCC representation. This used 40 cepstral coefficients, a short-time Fourier transform window size of 128 samples, a hop length of 20 samples, and 64 Mel bins spanning 20 Hz to 800 Hz. The second was a gammatone representation, computed using a 64-channel auditory-inspired filterbank with a 40 ms analysis window, a 10 ms hop, and a minimum frequency of 20 Hz. MFCC-based and other time-frequency representations are widely used in heart sound analysis, and recent benchmarking studies show that the choice of representation can affect classification performance [16,36,45]. These parameter settings were chosen to describe time-frequency structure within the retained heart sound frequency range while keeping the input size computationally manageable. Both representations convert one-dimensional heart sound waveforms into two-dimensional time-frequency representations that can be used by image-based deep learning backbones. They also emphasise spectral structure differently, allowing the choice of representation itself to be evaluated. At a high level, the two representations used in this study can be expressed as follows:

$$P_\tau(k) = |X_\tau(k)|^2$$

where $X_\tau(k)$ is the short-time Fourier transform of the waveform at frame τ , and $P_\tau(k)$ is the corresponding power spectrum.

$$M_m(\tau) = \sum_k H_m(k) P_\tau(k), \quad m = 1, \dots, 64$$

where $H_m(k)$ is the m -th Mel filter, and $M_m(\tau)$ is the Mel-filterbank energy at frame τ .

$$\widetilde{M}_m(\tau) = \text{dB}(M_m(\tau))$$

where $\text{dB}(\cdot)$ denotes decibel scaling of the Mel spectrogram.

$$c_n(\tau) = \sum_{m=1}^{64} D_{n,m} \widetilde{M}_m(\tau), \quad n = 0, \dots, 39$$

where $D_{n,m}$ denotes the orthonormal DCT-II weight used to obtain the n -th MFCC coefficient from the m -th dB-scaled Mel value, and $c_n(\tau)$ is the n -th coefficient at frame τ . In this study, the MFCC representation used 40 retained coefficients, 64 Mel bins, an analysis window size of 128 samples, and a hop length of 20 samples.

For the gammatone representation, the waveform was passed through a 64-channel auditory-inspired filterbank and summarised over successive analysis windows:

$$y_c(t) = (x * g_c)(t), \quad c = 1, \dots, 64$$

where $x(t)$ is the input waveform, $g_c(t)$ is the impulse response of the c -th gammatone filter, and $y_c(t)$ is the corresponding filter output.

$$G_c(\tau) = \sqrt{\frac{1}{W} \sum_{t \in W_\tau} y_c(t)^2}$$

where W_τ denotes the analysis window at frame τ , W is the number of samples in that window, and $G_c(\tau)$ is the framewise gammatone magnitude. In this study, the gammatone representation used 64 channels, a 40 ms analysis window, a 10 ms hop, and a minimum frequency of 20 Hz.

These equations summarise the transformations conceptually rather than every implementation step. In practice, the MFCC representation was generated using the MFCC implementation provided by the torchaudio library, which computed MFCCs from a dB-scaled Mel spectrogram using an orthonormal DCT-II. The gammatone representation was generated using the gtgram function from the gammatone package, which summarised gammatone-filterbank energy over successive analysis windows. The exact representations followed the software implementations and parameter settings described above.

The resulting MFCC and gammatone representations were resized to square images and replicated across three channels to match the input format expected by the pretrained vision backbones used in this study. These representations were then normalised using the mean and standard deviation computed from the corresponding training split. The same statistics were applied to the validation and test data so that preprocessing did not use information from the held-out data. Image size depended on the backbone family and was chosen to match the pretrained model variant. MobileNetV2, MobileNetV3-Large, and EfficientNet-B0 used 224×224 inputs. SwinV2-Tiny and SwinV2-Small used 256×256 inputs. EfficientNetV2-S used 384×384 inputs.

2.5. Deep learning backbones

Six deep learning backbones were evaluated. These were MobileNetV2, MobileNetV3-Large, EfficientNet-B0, EfficientNetV2-S, SwinV2-Tiny, and SwinV2-Small. All models were instantiated using the timm library with ImageNet-pretrained weights, three-channel input, and a single output logit for binary classification. Pretrained weights were used to support transfer learning rather than training all backbones from scratch on the dataset used in this study.

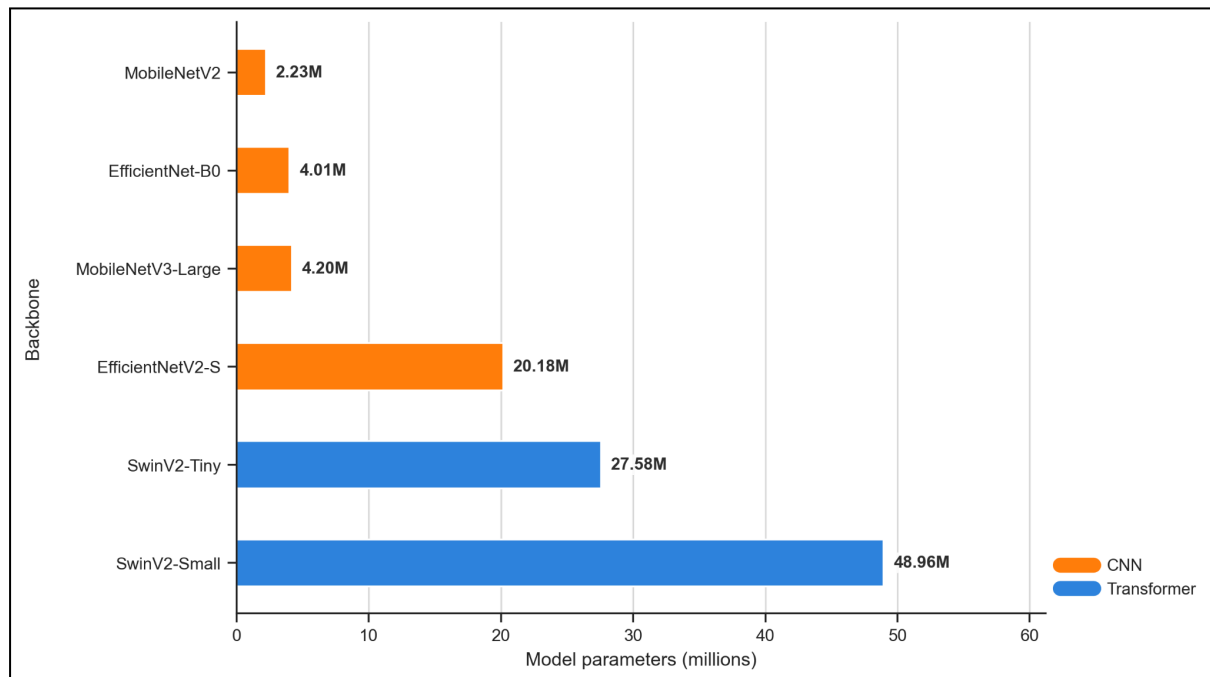


Figure 4. Parameter sizes of the evaluated deep learning backbones.

These backbones were chosen to span both lightweight CNNs and hierarchical vision transformers. This allowed comparison between compact CNN architectures, which learn mainly through local convolutional filters, and transformer-based models, which use attention mechanisms to model relationships across image regions. The selected backbones also covered a range of parameter counts, allowing the effect of model size to be considered alongside architectural differences.

2.6. Data splitting and experimental design

All partitioning of train, validation, and test sets was performed at the patient level. This was necessary because each patient contributed multiple recordings across different devices and auscultation sites. This ensured that recordings from the same patient could not appear in both the training or validation sets and the held-out test set.

A held-out patient-level split was created using stratified sampling on the binary reduced-LVEF label, with 65% of patients assigned to training, 15% to validation, and 20% to testing. These proportions kept most patients available for model training while still reserving validation and test cohorts large enough for threshold tuning and reasonably stable held-out performance estimates, particularly for the minority reduced-LVEF group. This split was used for the final within-device, cross-device, and pooled-device experiments.

Within-device models were trained on source-device recordings from the training split, monitored on the validation split, and evaluated on the same-device recordings from the held-out test split. Cross-device experiments also used the held-out test split, but evaluation was restricted to recordings from different target devices. Pooled-device evaluation used the pooled held-out test split across all devices. The F1 score for the positive class guided threshold tuning, early stopping, and configuration ranking throughout the study.

2.6.1. Within-device experiments

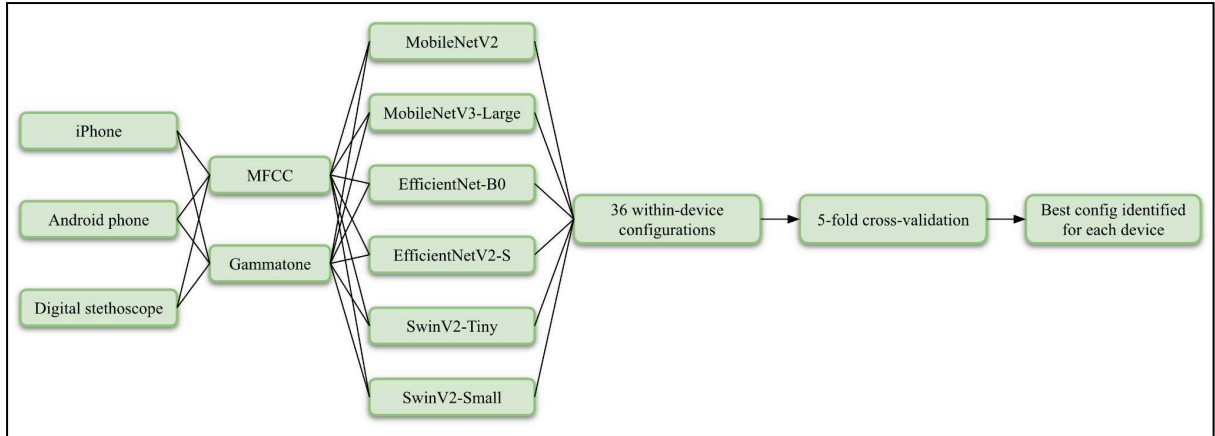


Figure 5. Within-device experiment workflow. For each training device, all combinations of two representations and six backbones were evaluated, yielding 36 within-device configurations. Model selection was based on five-fold cross-validation, after which the best configuration was identified for each device.

Within-device experiments were performed for iPhone, Android phone, and digital stethoscope recordings. For each device, MFCC and gammatone representations were paired with all six backbones, yielding 12 candidate configurations per device and 36 within-device configurations in total across the three devices. Model selection used a separate stratified five-fold patient-level cross-validation scheme. This cross-validation scheme was used only for within-device configuration ranking and not for the final evaluation on the held-out test split. In each cross-validation run, one fold was used for testing, and 10% of the remaining patients were assigned to the validation set.

After model selection, the best configuration was identified for each device. The final best-config within-device model was then trained on source-device recordings from the 65% training cohort, with performance monitored on the 15% validation cohort and evaluated on the same-device recordings from the 20% test cohort.

2.6.2. Cross-device experiments

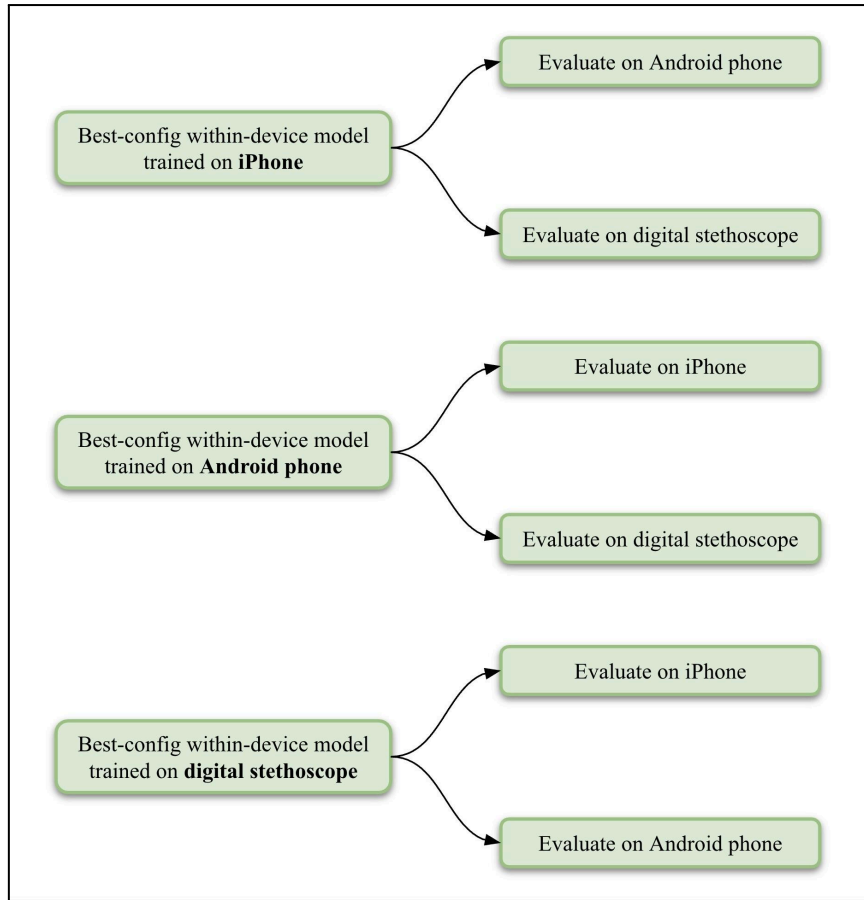


Figure 6. Cross-device experiment workflow. Each final best-config within-device model was evaluated on recordings from the two devices not used for training, yielding six pairwise cross-device transfer evaluations in total.

The cross-device experiments were designed to test domain generalisation. For each source device, the final best-config within-device checkpoint was reused in evaluation-only mode and applied to held-out recordings from the other two devices. This produced six pairwise transfer evaluations in total.

In each case, the source-device model was applied directly to target-device recordings with no additional training on the target device. This assessed how well a model trained in one recording domain transferred to a different recording domain. This experiment quantified the effect of device-related domain shift on held-out performance.

2.6.3. Pooled-device experiments

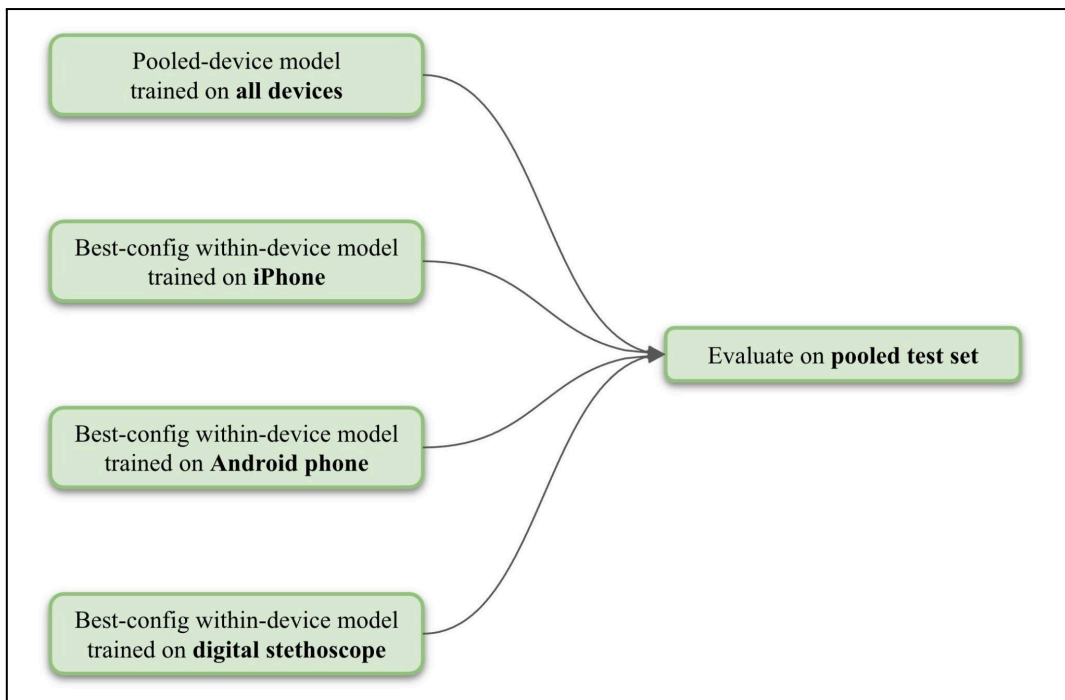


Figure 7. Pooled-device experiment workflow. The pooled-device model trained on all devices, together with the three final best-config within-device models, was evaluated on the same pooled test set for direct comparison.

A pooled-device setting was also evaluated to test whether combining recordings from all devices during training improved performance on mixed-device held-out data. The pooled-device configuration was not chosen by a separate pooled cross-validation scheme. Instead, one representation-backbone pairing was selected from the overall pattern observed in the within-device experiments. Backbone choice followed the architecture that performed most consistently well across devices, and representation choice followed the representation preferred by the majority of devices. This pooled-device configuration was used to train a pooled-device model.

The pooled-device model was evaluated on the pooled test set and compared against the three final best-config within-device models on that same pooled test set. This comparison separated the effect of training on multi-device data from the effect of training on a single recording domain.

2.7. Model training procedure

Models were trained with binary cross-entropy loss with logits, implemented in PyTorch as BCEWithLogitsLoss. Because reduced LVEF was the minority class, mistakes on reduced-LVEF recordings were penalised more heavily during training than mistakes on non-reduced-LVEF recordings. This was implemented through the pos_weight argument in BCEWithLogitsLoss. The size of this penalty was determined from the class balance in the training data. Specifically, it was calculated as the number of non-reduced-LVEF recordings divided by the number of reduced-LVEF recordings. This value was recalculated for each training subset because the class balance could vary slightly from one split to another.

Shared training settings are summarised in Table 3. Models were trained with BCEWithLogitsLoss, AdamW, an initial learning rate of 1.00×10^{-4} , weight decay of 1.00×10^{-4} , and a batch size of 32. Cosine annealing was used as the learning rate scheduler, with a minimum learning rate of 1.00×10^{-6} and five warmup epochs. Early stopping was triggered after 15 epochs without improvement in validation F1 for the positive class. Mixed precision was enabled during GPU training to reduce memory use and improve training speed.

Setting	Specification
Loss function	BCEWithLogitsLoss
Optimizer	AdamW
Initial learning rate	1.00×10^{-4}
Weight decay	1.00×10^{-4}
Batch size	32
Learning rate scheduler	Cosine annealing
Minimum learning rate	1.00×10^{-6}
Warmup epochs	5
Early stopping	Patience 15 epochs

Table 3. Shared training settings used across experiments.

At the end of each epoch, validation performance was computed. Thresholds from 0.05 to 0.95 in steps of 0.05 were evaluated on the validation set, and the threshold that maximised positive class F1 was selected. The best checkpoint was defined by the validation F1 score and stored together with its tuned threshold. Final test evaluation always used the best saved checkpoint rather than the last training epoch. All training, evaluation, and reporting experiments were conducted in Google Colab using NVIDIA A100 Tensor Core GPU resources.

2.8. Performance evaluation and uncertainty estimation

F1 for the positive class was used throughout the study for threshold tuning, early stopping, and configuration ranking. This choice was motivated by class imbalance and by the need to balance sensitivity and precision when identifying models for reduced-LVEF detection. In addition to F1, sensitivity, specificity, and AUROC were also reported.

F1, sensitivity, and specificity were calculated using the tuned validation threshold stored with each saved checkpoint. This threshold was carried forward to the held-out evaluation and was not tuned on the test set. In the pooled test comparison, the pooled-device model and the three final best-config within-device models were compared using their own tuned thresholds. AUROC was also reported as a threshold-free measure of discrimination.

For uncertainty estimation, patient-cluster bootstrap resampling was used. Bootstrap samples were created by resampling patients with replacement. When a patient was selected, all recordings from that patient were kept together in the bootstrap sample. This was important because multiple recordings from the same patient are related and should not be treated as completely separate data points when estimating confidence intervals. If recordings had been resampled one by one instead, the resulting confidence intervals could have been overly optimistic. A total of 2,000 bootstrap replicates were used. This was chosen as a practical compromise between the stability of the bootstrap confidence intervals and computational cost. Patient-cluster bootstrap confidence intervals were reported for pooled-test threshold-dependent metrics (F1, sensitivity, and specificity), AUROC relative to the random baseline, and auscultation-site-level pooled-device model AUROC.

3. Results

3.1. Within-device model selection

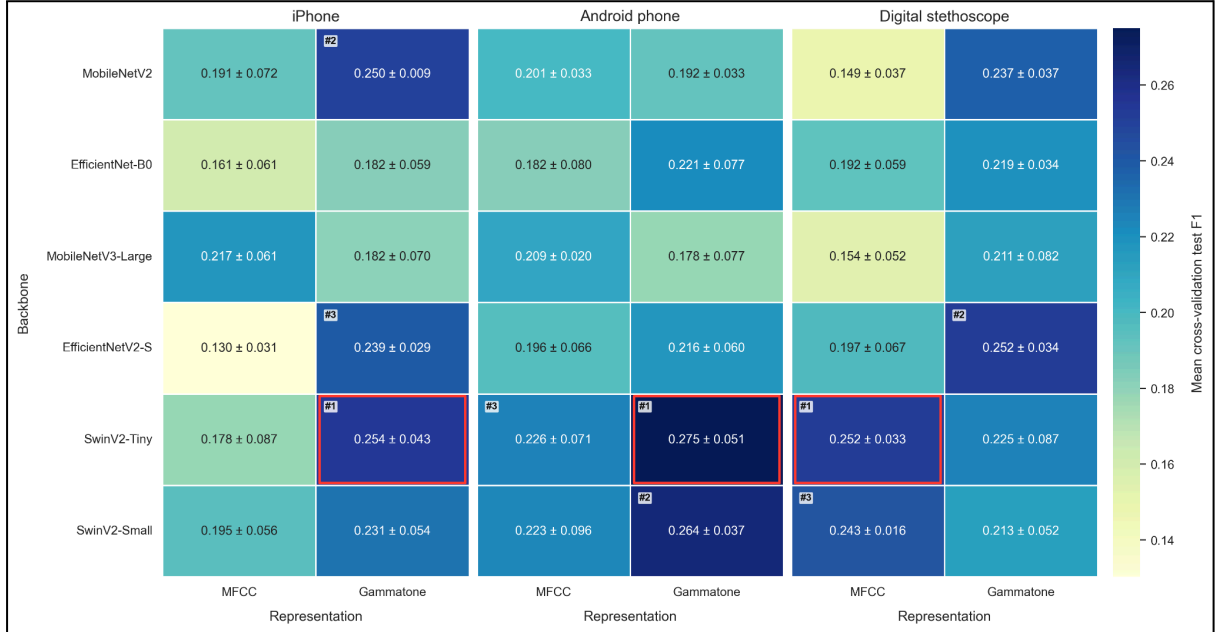


Figure 8. Cross-validation heatmaps for within-device model selection. Each heatmap shows the mean cross-validation test F1 score for all representation-backbone combinations for a training device. Cells are annotated with mean $\pm$ standard deviation across the five folds. The red outline marks the selected best configuration, and #1, #2, and #3 denote the three highest-performing configurations for that device.

Within-device cross-validation identified different best configurations for the three training devices. For iPhone recordings, the selected configuration was SwinV2-Tiny with gammatone, with a mean cross-validation test F1 of 0.254 ± 0.043 . For Android phone recordings, the selected configuration was also SwinV2-Tiny with gammatone, with a mean cross-validation test F1 of 0.275 ± 0.051 . For digital stethoscope recordings, the selected configuration was SwinV2-Tiny with MFCC, with a mean cross-validation test F1 of 0.252 ± 0.033 .

The top-ranked configurations were concentrated among a small subset of models. On iPhone data, the three highest-performing configurations were SwinV2-Tiny with gammatone, MobileNetV2 with gammatone, and EfficientNetV2-S with gammatone. On Android phone data, the three highest-performing configurations were SwinV2-Tiny with gammatone, SwinV2-Small with gammatone, and SwinV2-Tiny with MFCC. On digital stethoscope data, the three highest-performing configurations were SwinV2-Tiny with MFCC, EfficientNetV2-S with gammatone, and SwinV2-Small with MFCC.

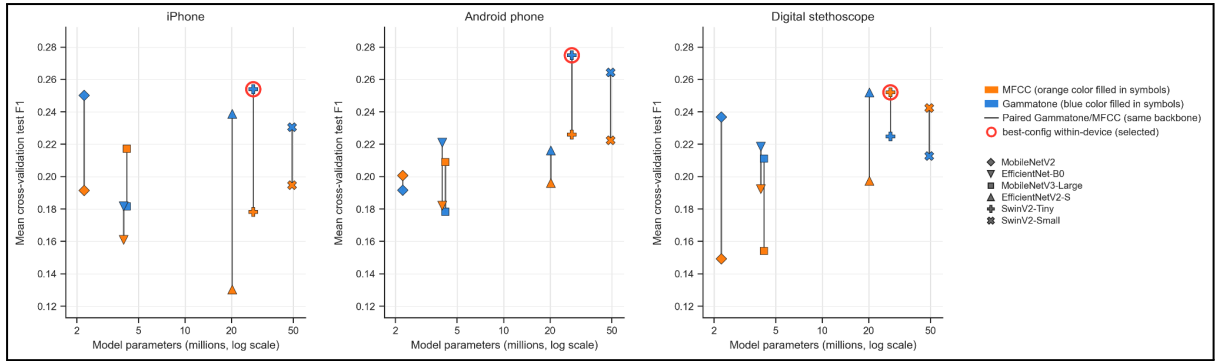


Figure 9. Cross-validation performance versus model size by training device. Each panel shows the mean cross-validation test F1 against backbone parameter size for one training device. Colours denote the representation (MFCC or Gammatone), vertical line segments link paired results from the same backbone, and the red outline marks the selected best configuration.

Across devices, the highest cross-validation F1 values were generally associated with the transformer backbones. SwinV2-Tiny and SwinV2-Small occupied the upper end of performance on Android phone and digital stethoscope data, while iPhone data also showed strong performance from MobileNetV2 with gammatone. The selected configuration on all three devices used SwinV2-Tiny. The largest backbone, SwinV2-Small, was not consistently the best-performing model.

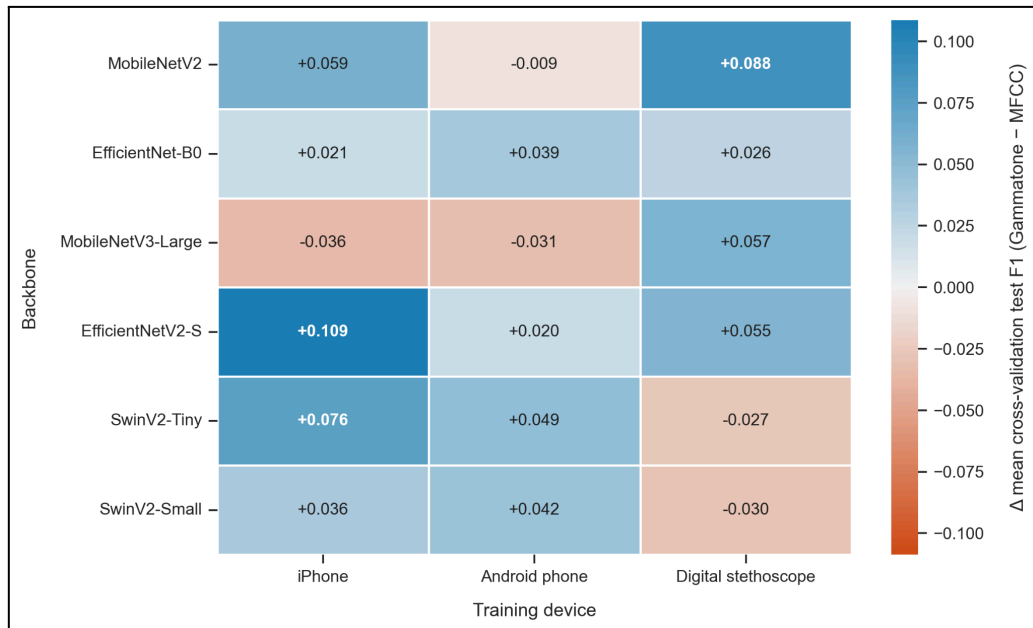


Figure 10. Cross-validation representation effect by training device and backbone. Each cell shows the difference in mean cross-validation test F1 between Gammatone and MFCC (Gammatone – MFCC) for a given training device and backbone. Positive values favour Gammatone, whereas negative values favour MFCC.

On iPhone data, gammatone achieved a higher mean cross-validation test F1 than MFCC for five of the six backbones, with the largest gain observed for EfficientNetV2-S

(+0.109). On Android phone data, gammatone also outperformed MFCC for four of the six backbones, with the largest gain observed for SwinV2-Tiny (+0.049). On digital stethoscope data, gammatone was favoured by the CNN backbones, with gains ranging from +0.026 to +0.088, whereas MFCC was favoured by the two Swin backbones, with differences of -0.027 for SwinV2-Tiny and -0.030 for SwinV2-Small.

After model selection, new models using the selected configurations were trained on the 65% training cohort, monitored on the 15% validation cohort, and evaluated on their own held-out same-device test sets. The resulting AUROC values were closely matched across devices, at 0.666 for iPhone, 0.664 for Android phone, and 0.665 for the digital stethoscope. These best-config within-device AUROC results are shown in the ROC curves in Section 3.2, providing the reference point for the cross-device experiments.

3.2. Cross-device performance

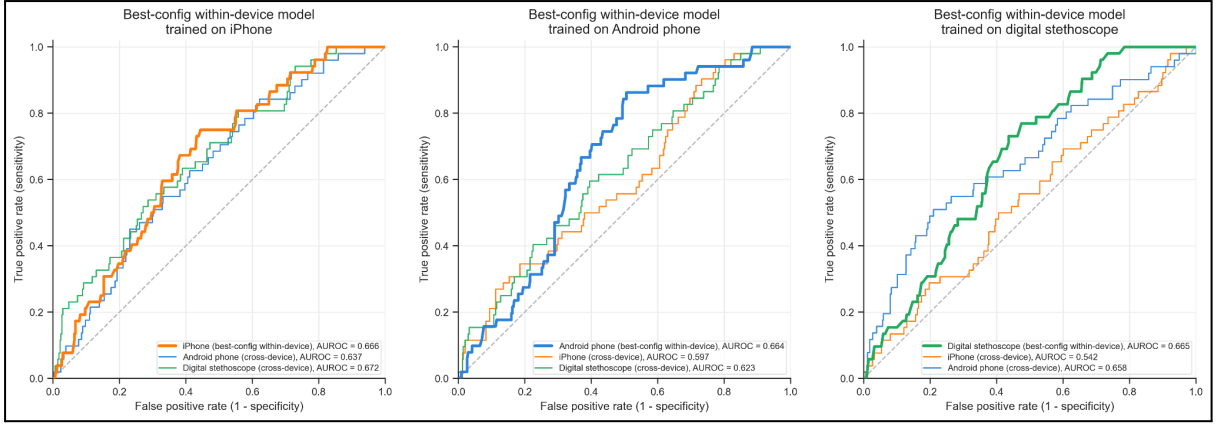


Figure 11. ROC curves for within-device and cross-device evaluation by source model. Each panel corresponds to the best-config within-device model trained on one device. The within-device ROC curve is shown together with the two cross-device ROC curves obtained by evaluating the source model on recordings from the other two devices.

The cross-device experiments used the three best-config within-device models identified in Section 3.1. These were SwinV2-Tiny with gammatone for iPhone, SwinV2-Tiny with gammatone for Android phone, and SwinV2-Tiny with MFCC for digital stethoscope.

Cross-device performance varied by transfer direction. Cross-device AUROC values ranged from 0.542 to 0.672. The highest cross-device AUROC was obtained when the iPhone trained model was evaluated on digital stethoscope recordings (0.672). The lowest cross-device AUROC was obtained when the digital stethoscope trained model was evaluated on iPhone recordings (0.542). The Android to iPhone direction was also lower, with an AUROC of 0.597.

3.3. Pooled-device comparison

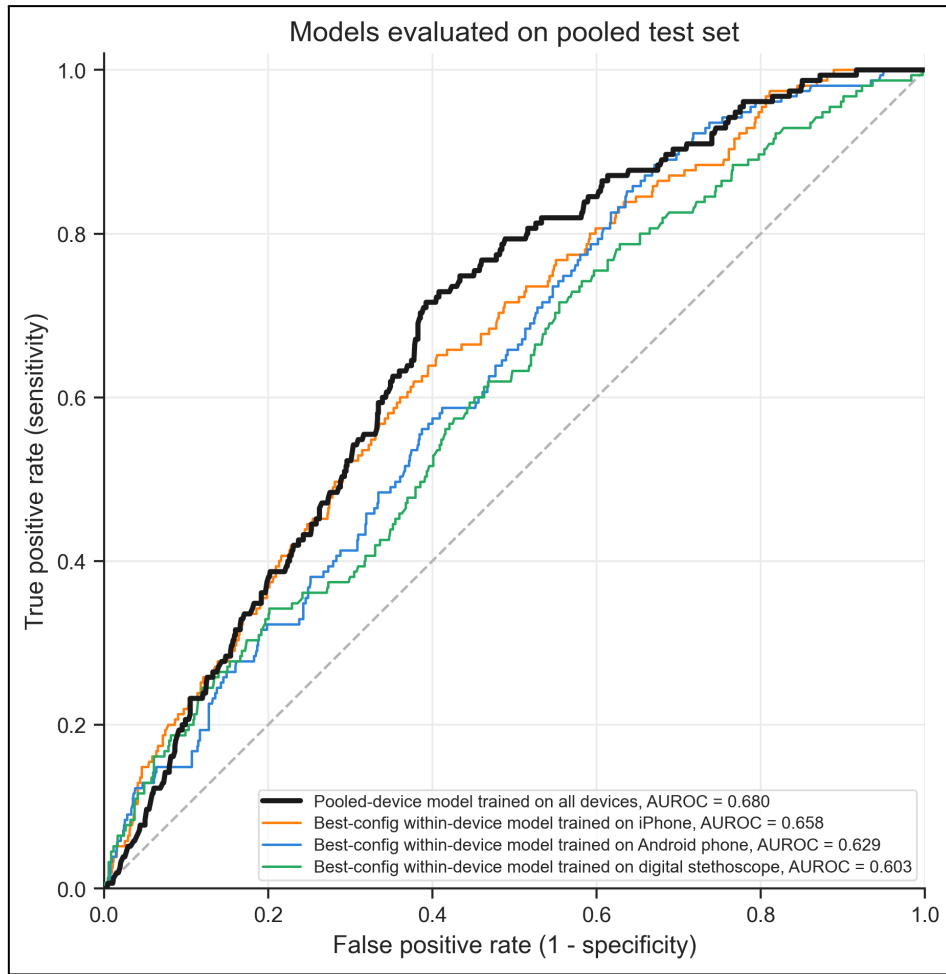


Figure 12. ROC comparison on the pooled test set. The pooled-device model trained on all devices is compared against the three best-config within-device models, each evaluated on the same pooled test set. This comparison assesses whether training on pooled multi-device data improves discrimination relative to device-specific training alone.

The pooled-device model used SwinV2-Tiny with gammatone. SwinV2-Tiny was selected because it was the backbone used by the best-config within-device model for all three devices, whereas gammatone was selected because it was preferred by two of the three devices.

On the pooled test set, the pooled-device model achieved the highest AUROC at 0.680. The three best-config within-device models evaluated on that same pooled test set had AUROC values of 0.658 for the iPhone trained model, 0.629 for the Android trained model, and 0.603 for the digital stethoscope trained model. The ROC comparison ranked the pooled-device model highest by AUROC in this pooled-test evaluation.

Model evaluated on pooled test set	Tuned threshold	F1, % (95% CI)	Sensitivity, % (95% CI)	Specificity, % (95% CI)
Pooled-device model trained on all devices	0.4	29.5 (17.4-40.4)	81.9 (74.7-89.1)	43.1 (37.3-49.0)
Best-config within-device model trained on iPhone	0.45	28.9 (16.9-39.4)	67.7 (56.1-79.5)	54.1 (48.7-59.7)
Best-config within-device model trained on Android phone	0.4	26.7 (15.4-37.4)	94.8 (91.1-98.1)	21.4 (17.2-26.2)
Best-config within-device model trained on digital stethoscope	0.45	24.4 (13.9-34.6)	98.1 (95.0-100.0)	7.7 (5.2-10.3)

Table 4. Pooled-test performance of the pooled-device model and best-config within-device models. Metrics were computed using each model's tuned threshold. 95% confidence intervals were estimated using the patient-cluster bootstrap resampling.

Threshold-dependent metrics on the pooled test set also ranked the pooled-device model highest by F1. The pooled-device model achieved the highest F1 score of 29.5% (95% CI 17.4% to 40.4%), with sensitivity of 81.9% (74.7% to 89.1%) and specificity of 43.1% (37.3% to 49.0%). The iPhone trained within-device model achieved an F1 score of 28.9% (16.9% to 39.4%), with lower sensitivity at 67.7% (56.1% to 79.5%) and higher specificity at 54.1% (48.7% to 59.7%). The Android trained within-device model achieved 94.8% sensitivity (91.1% to 98.1%) but had lower specificity at 21.4% (17.2% to 26.2%), with F1 reduced to 26.7% (15.4% to 37.4%). The digital stethoscope trained within-device model showed the highest sensitivity at 98.1% (95.0% to 100.0%) and the lowest specificity at 7.7% (5.2% to 10.3%), with the lowest F1 at 24.4% (13.9% to 34.6%).

3.4. Robustness relative to the random baseline

Model and evaluation setting	AUROC – 0.5 (95% CI)	95% CI above AUROC baseline?
Best-config within-device model trained on iPhone (same-device test set)	0.166 (0.064-0.261)	Yes
Best-config within-device model trained on Android phone (same-device test set)	0.164 (0.084-0.239)	Yes
Best-config within-device model trained on digital stethoscope (same-device test set)	0.165 (0.086-0.239)	Yes
Cross-device: iPhone → Android phone	0.137 (0.050-0.227)	Yes
Cross-device: iPhone → Digital stethoscope	0.172 (0.062-0.284)	Yes
Cross-device: Android phone → iPhone	0.097 (–0.012-0.191)	No
Cross-device: Android phone → Digital stethoscope	0.123 (0.017-0.219)	Yes
Cross-device: Digital stethoscope → iPhone	0.042 (–0.083-0.167)	No
Cross-device: Digital stethoscope → Android phone	0.158 (0.065-0.246)	Yes
Pooled-device model trained on all devices (pooled test set)	0.180 (0.111-0.248)	Yes
Best-config within-device model trained on iPhone (pooled test set)	0.158 (0.080-0.242)	Yes
Best-config within-device model trained on Android phone (pooled test set)	0.129 (0.053-0.197)	Yes
Best-config within-device model trained on digital stethoscope (pooled test set)	0.103 (0.026-0.178)	Yes

Table 5. AUROC improvement above the random baseline of 0.5 with 95% patient-cluster bootstrap confidence intervals for each evaluation setting.

All three final within-device models remained above the random AUROC baseline, as indicated by the 95% patient-cluster bootstrap confidence intervals. The AUROC improvement above the baseline was 0.166 (95% CI 0.064 to 0.261) for the iPhone trained model, 0.164 (0.084 to 0.239) for the Android trained model, and 0.165 (0.086 to 0.239) for the digital stethoscope trained model.

Among the six cross-device evaluations, four remained above the random AUROC baseline with bootstrap confidence intervals. The largest AUROC improvement above baseline was observed for the iPhone to digital stethoscope transfer at 0.172 (95% CI 0.062 to 0.284). The two transfer directions into the iPhone were less robust. Android phone to iPhone showed an AUROC improvement of 0.097 with a 95% confidence interval from –0.012 to 0.191, and digital stethoscope to iPhone gave an AUROC improvement of 0.042 with a 95% confidence interval from –0.083 to 0.167.

All four pooled-test evaluations also remained above the random AUROC baseline. The pooled-device model showed the largest AUROC improvement above baseline at 0.180 (95% CI 0.111 to 0.248). The three best-config within-device models evaluated on the pooled test set had AUROC improvements above baseline of 0.158 (0.080 to 0.242) for the iPhone trained model, 0.129 (0.053 to 0.197) for the Android trained model, and 0.103 (0.026 to 0.178) for the digital stethoscope trained model.

3.5. Auscultation-site analysis

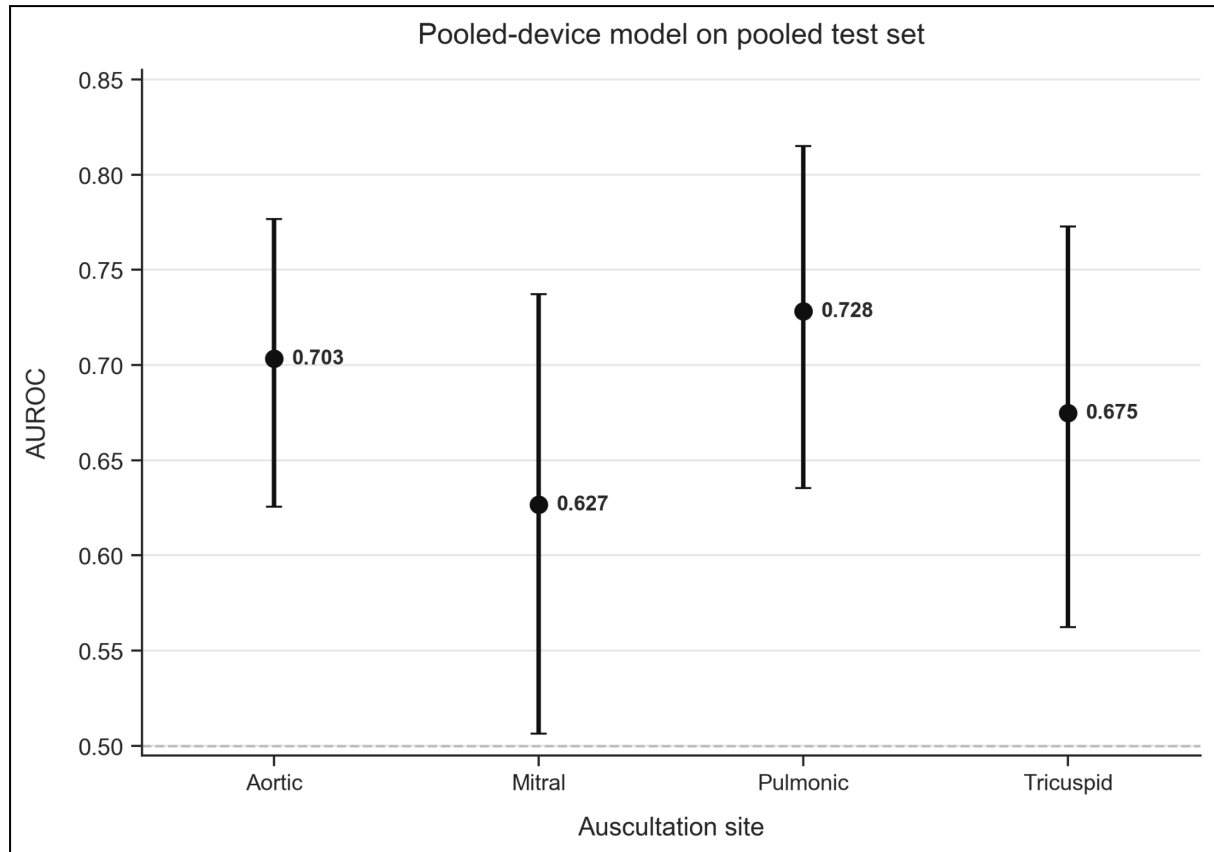


Figure 13. AUROC of the pooled-device model by auscultation site on the pooled test set. Points and error bars show recording-level AUROC with 95% patient-cluster bootstrap confidence intervals for each auscultation site.

The pooled-device model showed different AUROC values across the four auscultation sites. Pulmonic recordings gave the highest AUROC at 0.728 (95% CI 0.635 to 0.815), followed by aortic recordings at 0.703 (0.626 to 0.777), tricuspid recordings at 0.675 (0.562 to 0.773), and mitral recordings at 0.627 (0.506 to 0.737). All four confidence intervals remained above the random AUROC baseline of 0.5.

4. Discussion

4.1. Overview of the main findings

This study examined whether deep learning models trained on PCGs recorded with smartphones and a digital stethoscope could detect reduced LVEF and whether performance remained stable across recording domains. The results suggest that reduced LVEF can be detected from these recordings, with AUROC values above the random baseline in the final within-device and pooled-device evaluations, but that performance changed markedly when models were evaluated on the recordings from different devices.

The within-device model selection identified SwinV2-Tiny as the strongest backbone across all three devices, although the preferred time-frequency representation differed by device. Gammatone was selected for the iPhone and Android phone recordings, whereas MFCC was selected for the digital stethoscope recordings. This indicates that the choice of representation was not universal across device types, even when the same backbone was favoured. The cross-device experiments showed that performance did not transfer equally across device pairs. The AUROC values were closely matched across the three final within-device models, whereas the cross-device AUROC values were more variable and included weaker transfer into the iPhone recordings. This pattern is consistent with device-related domain shift and supports the expectation that cross-device performance would be less stable than within-device performance. The pooled-device model achieved the strongest overall performance on the pooled test set, with the highest AUROC and the highest F1 among the models compared. The bootstrap analysis further showed that the final within-device and the pooled-test AUROC values remained above the random baseline, while cross-device robustness was less consistent. These findings therefore support the two expectations set out in Section 1.5. The cross-device performance was less stable than the within-device performance, and the pooled-device training gave the strongest pooled-test performance by point estimate. In addition, the analysis by auscultation site suggested that the pooled-device discrimination was highest at the pulmonic site and remained comparatively strong at the aortic site.

Overall, these findings support the view that PCG-based reduced-LVEF screening may be feasible. However, performance dropped when the models trained on a single recording domain were evaluated on the recordings from different devices. This suggests that training strategies that address device-related domain shift are important.

4.2. Within-device model selection and performance

One of the clearest within-device findings was the repeated selection of SwinV2-Tiny. Although several backbones performed well on individual devices, SwinV2-Tiny was selected for the iPhone, Android phone, and digital stethoscope recordings. The smallest CNN backbone was not the top performer. The largest model, SwinV2-Small, also did not consistently outperform the others. Instead, the strongest within-device results repeatedly came from SwinV2-Tiny, a hierarchical transformer that uses the attention mechanism while remaining smaller than SwinV2-Small. This suggests that the task benefited from that particular combination of architecture and model size. The fact that SwinV2-Small did not consistently outperform SwinV2-Tiny further suggests that simply increasing the number of parameters was not sufficient to improve the within-device performance. One possible explanation is that the larger model may have been more prone to overfitting, so the additional parameters did not translate into better held-out performance.

The representation results also depended on the device. Gammatone was favoured on both smartphone domains, whereas MFCC was favoured for the selected digital stethoscope model. This suggests that the choice of representation may have been influenced by differences in how each device captured and recorded heart sounds [17,19,20]. The representation that worked best on one device did not necessarily work best on another. These results therefore suggest that MFCC and gammatone should be compared separately for each recording domain, rather than assuming that one representation will generalise best across all recording domains.

The overall level of within-device performance is easier to interpret when compared with earlier PCG studies. Deep learning studies on murmur detection and structural heart disease screening have often reported stronger performance [23,24], which is plausible because those targets have more direct acoustic features [15,16]. By contrast, reduced LVEF is an echocardiographic measure of ventricular function [1,3,4], and phonocardiographic and acoustic cardiography studies suggest that related heart sound differences are present but indirect [7-12]. Against that background, the present results are important not because they match murmur-level performance, but because they show that PCGs alone still contain enough information for the above-random reduced-LVEF discrimination.

4.3. Cross-device performance and device-related domain shift

The cross-device experiments provided clear evidence that device-related domain shift is a central problem for this task. The final within-device AUROC values were tightly clustered, yet the cross-device AUROC varied noticeably by the transfer direction. A model that performed well on recordings from one device did not always perform similarly on recordings from another device. The weaker transfer to the iPhone recordings was notable, showing that the performance was not uniform across all transfer directions.

This pattern is plausible because PCGs are not acquired in the same way across recording domains. Recording devices can differ in microphone response, how the device contacts the chest, app-level signal processing, and susceptibility to environmental noise [17,19,20]. These differences can alter the spectral and temporal profiles of the recording, even when the same physiological event is being measured [19,20,32,34]. The present findings therefore reinforce a methodological concern already raised in multi-device PCG research, namely that recordings from different devices should not be treated as automatically interchangeable [32,34]. The current study strengthens this point by showing it directly in the reduced-LVEF setting. Strong within-device validation can coexist with weaker cross-device transfer if part of what the model learns is tied to device-specific signal patterns.

The cross-device results are still informative in a positive sense. Not all transfer directions failed. Four of the six cross-device AUROC estimates remained above the random baseline according to the bootstrap confidence intervals, which suggests that at least part of what the models learned was genuinely related to reduced LVEF rather than purely to device artefact. However, the inconsistency across the transfer directions makes it difficult to justify single-device training as a sufficient strategy for real-world clinical use across different recording domains. For smartphone-oriented screening in particular, this matters because, in practice, deployment involves many device types rather than one standardised recording platform [17,19-22].

4.4. Pooled-device training and pooled-test performance

The pooled-device results suggest that training on recordings from multiple devices can improve robustness in this setting. When evaluated on the pooled test set, the pooled-device model achieved the highest AUROC and the highest F1 among the models compared. This is notable because the pooled-device configuration was not chosen through a separate pooled cross-validation procedure. Instead, it was selected based on the overall pattern observed in the within-device experiments. Even with that pragmatic selection strategy, the pooled-device model gave the strongest pooled-test performance by point estimate.

The F1, sensitivity, and specificity results help clarify why the pooled-device model is important. The pooled-device model not only improved AUROC. It also showed a more balanced trade-off between sensitivity and specificity than the within-device models. The Android trained and digital stethoscope trained models retained very high sensitivity on the pooled test set, but this came with a substantial loss of specificity. By contrast, the pooled-device model preserved high sensitivity while retaining more specificity, which yielded the strongest F1 score overall. In a screening or triage setting, this balance matters because very low specificity would lead to more false-positive referrals for confirmatory echocardiography, whereas very low sensitivity would miss patients who may benefit from further cardiac evaluation.

These pooled-device findings fit naturally with the cross-device results. If performance weakens when models trained on one device are transferred to another, then pooled-device training is a logical way to reduce that problem by exposing the model to a broader range of signal conditions during learning. While the present study does not prove that pooled-device training is the only or best solution to domain shift, it shows that pooled-device training performed better than single-device training in the pooled evaluation setting tested in this study.

4.5. Clinical and methodological implications

From a clinical perspective, the results support the feasibility of PCG-based reduced-LVEF screening in this dataset rather than demonstrating a clinically established triage tool. The task itself is challenging because reduced LVEF is a functional phenotype [1,3,4] rather than a directly audible acoustic target, and older phonocardiographic and acoustic cardiography studies suggest that its heart sound associations are indirect [7-12]. A more realistic interpretation is that smartphone and digital stethoscope PCGs may contain signals that are relevant to reduced-LVEF screening and may help identify individuals who could benefit from further cardiac evaluation. This view is consistent with adjacent literature on artificial intelligence (AI)-enabled cardiac auscultation, including murmur detection [23,24] and ECG-enabled stethoscope screening for reduced LVEF [18,30], where biosignal analysis is used to support referral or screening workflows rather than replacing confirmatory imaging. The present study follows the same screening logic, but applies it to reduced LVEF using PCGs alone.

The present findings also extend that literature in a more specific way. The studies most directly related to this task remain murmur or structural disease models [23,24], ECG-enabled or multimodal reduced-LVEF studies [18,29,30], and a single-platform digital stethoscope foundation model study that included PCG-only reduced-LVEF modelling [31]. None of these studies directly evaluates PCG-only reduced-LVEF modelling across within-device, cross-device, and pooled-device settings [18,23,24,29-31]. This distinction matters because a model that performs well within one recording domain may not generalise reliably to another, making cross-device evaluation essential for judging practical applicability [32,34]. The novelty of the present study therefore lies not only in applying deep learning to this more challenging functional target but also in comparing within-device, cross-device, and pooled-device evaluations within the same overall framework. Demonstrating the above-random within-device and pooled-device performance under these conditions supports the view that PCGs alone can carry information relevant to reduced LVEF in this dataset, while also showing that device-related domain shift is a practical barrier to generalisation. If only within-device performance had been reported, the results would have given a more optimistic picture of model behaviour. The cross-device experiments showed that this would have been incomplete, which is consistent with the concerns in PCG modelling about domain variability and transfer across recording conditions [32,34].

The auscultation-site analysis adds another useful point. The highest pooled-device AUROC was observed at the pulmonic site, with the aortic site next highest. This may indicate that the recordings from these positions contained more useful information for the reduced-LVEF discrimination in the current dataset. One possible explanation is that the aortic and pulmonic areas may have captured outflow-related acoustic information more clearly, so subtle changes linked to impaired systolic function may have been easier to detect there. However, this finding should be interpreted cautiously. Site-level differences could also reflect physiology, recording quality, positioning consistency, or interactions with device-specific acoustics. It would therefore be premature to treat the present ranking of sites

as a basis for changing acquisition practice. Even so, the results suggest that the auscultation site is not just a label in the dataset and may need to be considered more explicitly in future PCG screening studies. A dedicated comparison of standard auscultation sites for PCG-based reduced-LVEF detection has not been clearly reported in prior published studies.

4.6. Limitations and future directions

Several limitations should be considered when interpreting these findings. First, the study used the de-identified subset from the parent clinical study that was not originally designed specifically for reduced-LVEF model development. The LVEF labels were derived from echocardiography performed within the preceding three years rather than from imaging obtained at the exact time of sound recording. This introduces a potential label mismatch, because ventricular function may have changed between echocardiography and PCG acquisition, and because LVEF measurement itself is not perfectly reproducible [25]. The dataset also contained no demographic or broader clinical covariates, so it was not possible to examine whether performance differed across ethnic groups, age, sex, comorbidity profile, or other patient factors.

Second, the data came from a single healthcare context, and all participants were adults. This limits the extent to which the results can be assumed to generalise to other populations, ethnic groups, age groups, or recording practices. The recordings were collected in real clinical environments rather than in acoustically controlled settings. This is a strength in terms of realism, but it also introduces additional variability. The Android recordings were obtained from two handset models, yet the de-identified dataset did not retain handset-level labels. As a result, this particular device-related variation could not be analysed in greater detail. In addition, a signal-quality annotation was not available, so this study could not determine whether the poor generalisation reflected subtle disease-related signal differences or inconsistent recording quality.

Third, the experimental scope was deliberately focused. The cross-device testing was performed only with the final best-config within-device models, not with all 36 candidate within-device configurations. This means that the study cannot rule out the possibility that a configuration that was suboptimal might have generalised better across devices. The pooled-device configuration was also selected pragmatically from the within-device pattern rather than through a dedicated pooled cross-validation search. Additionally, only two time-frequency representations and six pretrained backbones were evaluated, and the shared training settings were not tuned separately for every configuration. Other input representations, models trained directly on waveforms, alternative architectures, or configuration-specific hyperparameter optimisation might behave differently.

These limitations point directly to the most useful next steps. Future work should test the present approach in external cohorts and in more diverse multi-device datasets, especially if smartphone deployment is the long-term goal. More exhaustive cross-device model

selection would be valuable because the configuration that is best within-device may not be the one that transfers best across recording domains. Patient-level aggregation should also be explored, since clinical decisions are ultimately made for patients rather than for individual recordings. However, that would require an explicit aggregation rule or a different learning framework, such as multiple instance learning (MIL). Furthermore, additional extensions such as multimodal PCG + ECG models, signal-quality annotation, and data augmentation could be combined with interpretability methods such as Gradient-weighted Class Activation Mapping (Grad-CAM) and SHapley Additive exPlanations (SHAP), which could help clarify which parts of the signal are genuinely informative and which are vulnerable to domain shift.

Overall, these limitations do not negate the main findings of this study. Instead, they clarify how those findings should be interpreted. The results support the feasibility of reduced-LVEF screening from PCGs alone while also showing that device-related domain shift is a practical obstacle that cannot be ignored. Future research should therefore focus not only on improving best-case performance, but also on improving robustness across devices, recording settings, and patient groups.

5. Conclusion

This study showed that deep learning models trained on PCGs recorded with smartphones and a digital stethoscope could detect reduced LVEF above the random baseline under within-device and pooled-device evaluation. Within-device model selection identified SwinV2-Tiny as the most consistent backbone across all three devices, although the preferred time-frequency representation differed by device. Cross-device experiments then showed that this performance did not transfer uniformly across devices, indicating that device-related domain shift is a practical challenge for this task. By contrast, the pooled-device model achieved the strongest overall performance by point estimate on the pooled test set, suggesting that training on recordings from multiple devices may improve robustness. A secondary finding was that the pooled-device performance also varied by auscultation site, with the pulmonic and aortic sites performing best, suggesting that recording location may influence model performance and merits further study in future PCG-based reduced-LVEF research.

These findings suggest that PCG-based reduced-LVEF screening may be feasible, even though reduced LVEF is a functional target rather than a directly audible acoustic target. A key contribution of this study was to compare within-device, cross-device, and pooled-device evaluation within one framework, showing that within-device success can coexist with weaker cross-device transfer. The main implication is not only that PCGs can carry information relevant to reduced-LVEF screening, but also that evaluation across recording domains is necessary if such deep learning models are intended for real-world clinical use. Future research should therefore focus on broader multi-device validation and on training strategies that explicitly address device-related domain shift.

Code availability

The code used in this study is available at <https://github.com/anywheredoor/phonocardiogram-lvef-deeplearning>.

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